

Image File cache for any reason, or if there is a transient network failure of sufficient duration to prevent a complete transfer to take place.

Once the entirety of a Software Image has been downloaded and is ready to apply, the OTA Requestor SHALL execute the "[Applying a software update](#)" sequence of the next section.

11.20.3.6. Applying a software update

Once a Software Image has been fully downloaded based on a [QueryImageResponse](#) command, the OTA Requestor SHALL proceed with a sequence to determine when to apply the update by invoking the [ApplyUpdateRequest](#) command.

UpdateToken usage

The OTA Requestor SHALL provide an [UpdateToken](#) to the OTA Provider with the [ApplyUpdateRequest](#) command. This token SHALL be a previously provided [UpdateToken](#) from the last [QueryImageResponse](#), unless the token was lost by the OTA Requestor. In case of token loss, the OTA Requestor SHALL use its [Operational Node ID](#) encoded as a 64-bit value in network byte order. This [UpdateToken](#) MAY be used by an OTA Provider to track deferred OTA application or otherwise allow short-term tracking of OTA Requestors for algorithmic bookkeeping. The OTA Provider SHALL NOT consider an invalid [UpdateToken](#) as a reason to continuously deny or delay an OTA Requestor's request to apply a Software Image.

Update application process

On receipt of the [ApplyUpdateRequest](#) command, the OTA Provider SHALL respond with an action to be taken by the OTA Requestor before activating the new version. The method used to determine the [Action](#) field of the response MAY be based on implementation-specific rules and logic.

Note that the [DelayedActionTime](#) field is a relative time delay from the moment of receipt, which needs to be computed by the OTA Provider to reflect the difference between the OTA Provider's current time and the desired time for execution of the Action.

In case of a successful invocation, the following actions to be taken by the OTA Requestor are possible, based on the [Action](#) field in the [ApplyUpdateResponse](#) command:

- Proceed: Apply the update, taking in account the delay time stated in [DelayedActionTime](#).
 - If the [DelayedActionTime](#) is zero, then the OTA Requestor SHALL apply the update without additional delay.
 - If the [DelayedActionTime](#) is non-zero, the OTA Requestor SHALL await at least [DelayedActionTime](#) seconds prior to applying the software update. An example use of this Action by an OTA Provider is to schedule application of a Software Image based on a user's preferred update time for Nodes of a certain type (e.g. light bulbs or window coverings) to occur at a time when the user is not at home, or when the temporary unavailability of the Node during the update would not pose a problem.
 - When the Proceed action is given, the OTA Requestor SHALL NOT invoke the [ApplyUpdateRequest](#) command again, unless the OTA Requestor suffers an error or unexpected condition while proceeding to apply the new Software Image.